

Seismic Activity Analysis Report

Interactive Pipeline Analysis of Global Earthquake Data

Data Range Analyzed: 2026-01-01 to 2026-01-29

Total Events Processed: 1140

Maximum Recorded Magnitude: 6.5 M

Generated By: Antigravity Pipeline Engine

Date: June 29, 2026

Executive Summary

This report contains analysis computed from data fetched dynamically from the **USGS Earthquake Hazards Program API**. A total of **1140** seismic events occurred during the selected window of **2026-01-01 to 2026-01-29**. The maximum recorded earthquake had a magnitude of **6.5 M**. The average magnitude across all tracked events is **4.54 M** with a median of **4.40 M**.

Dataset Descriptive Statistics

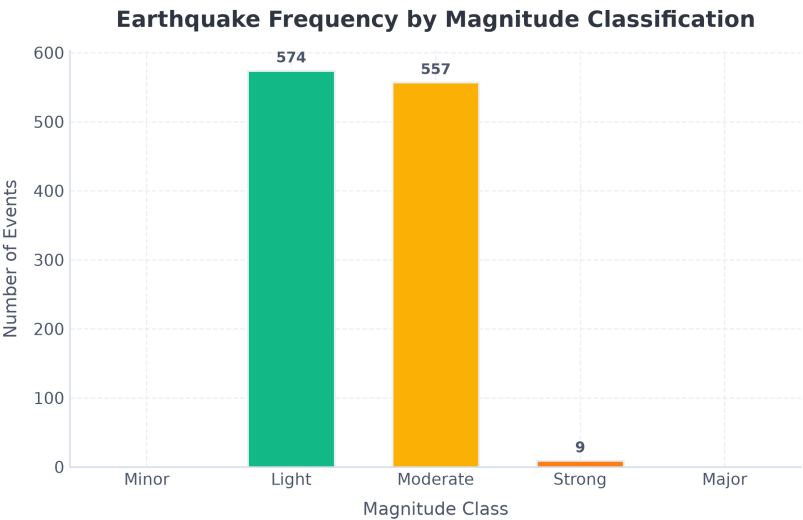
Below is a statistical breakdown of magnitude and depth indicators across all processed features.

Metric	Magnitude (M)	Depth (km)
Count	1140	1140
Mean (Average)	4.538	81.22
Median	4.40	0.00
Minimum Value	4.00	2.29
Maximum Value	6.50	638.18
Standard Deviation	0.395	0.00

Tsunami Warning Analysis: Out of 1140 earthquakes, **13** events triggered a tsunami warning (1.14% of all events). The mean magnitude for events triggering tsunami warnings was **5.37 M**, compared to **4.53 M** for non-tsunami events. This highlights the strong relationship between earthquake magnitude and ocean displacement risk.

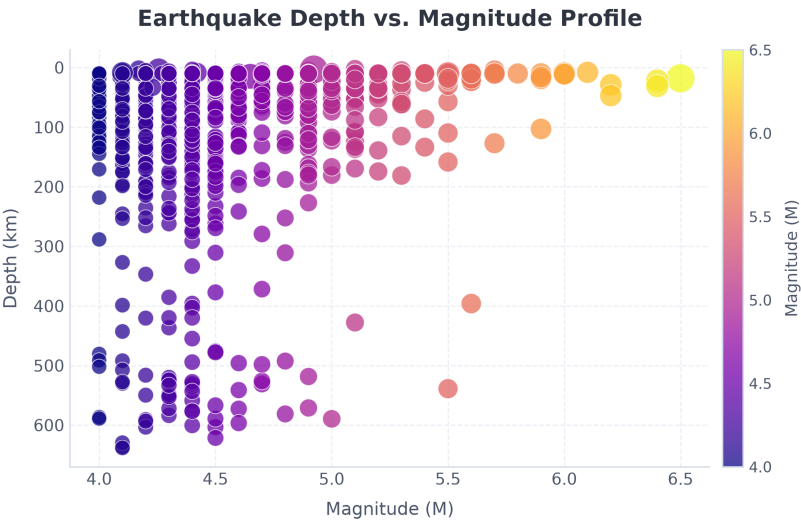
Magnitude Distribution

The chart below aggregates occurrences grouped by magnitude categories. Light and Minor tremors make up the vast majority of activities, which aligns with standard seismic frequency models.



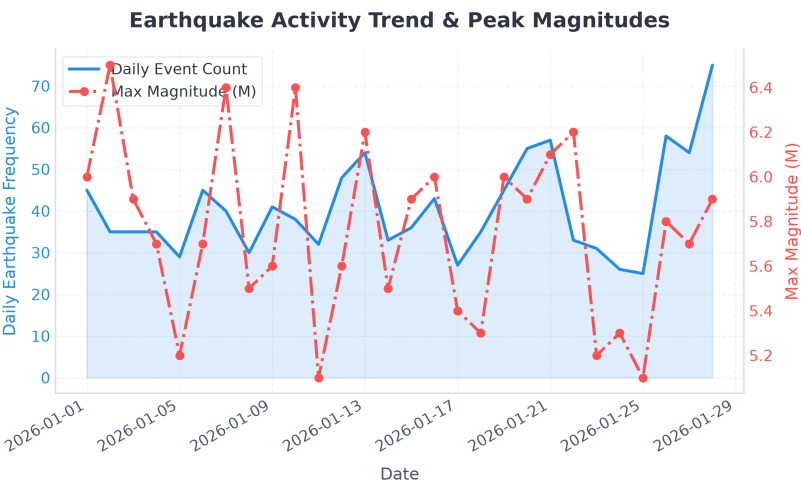
Depth and Physical Profiles

Analyzing how earthquake focal depth corresponds with event magnitude allows researchers to understand fault-line friction dynamics. Deeper earthquakes typically occur at subduction zones.



Statistical Analysis: The correlation between magnitude and depth is **-0.1373**. This indicates **no linear correlation** between an earthquake's magnitude and how deep it is recorded within the crust.

Chronological Trends & Max Energy Peak



Regional and Event Breakdowns

Top 5 Most Active Regions

Region / Location	Event Count	Max Magnitude (M)
Russia	133	6.2
Philippines	131	6.4
Indonesia	131	5.7
Japan	72	5.7
Tonga	41	5.5

Top 5 Strongest Seismic Events Recorded

Date & Time (UTC)	Magnitude (M)	Location	Depth (km)	Felt Reports
2026-01-02 13:58:15	6.5	0 km W of San Marcos, Mexico	18.0	528
2026-01-10 14:58:23	6.4	247 km SE of Sarangani, Philippines	31.0	13
2026-01-07 03:02:53	6.4	35 km E of Santiago, Philippines	22.0	36
2026-01-13 07:34:07	6.2	135 km SE of Kuril'sk, Russia	29.0	0
2026-01-22 12:42:34	6.2	127 km S of Vilyuchinsk, Russia	47.0	0

Pipeline Flow Verification

This PDF report marks the final step in the pipeline. The flow verified for this dataset consists of: (1) Fetching from USGS JSON API endpoint, (2) Clean & Flat-mapping using Pandas, (3) Generating visualizations with Matplotlib, (4) Statistical calculations, and (5) Exporting to PDF (ReportLab) & JSON/CSV files locally.